

1. Administrative & Study Identification

| Field | Data Entry / Details | | :--- | :--- | | **Full Study Title** | A Phase 3, Randomized, Study of Neoadjuvant Chemotherapy Alone Versus Neoadjuvant Chemotherapy Plus Nivolumab or Nivolumab and BMS-986205, Followed by Continued Post- Surgery Therapy With Nivolumab or Nivolumab and BMS-986205 in Participants With Muscle- Invasive Bladder Cancer | | **Short Title/Acronym** | ENERGIZE | | **Protocol Number** | CA017-078 | | **NCT Number** | NCT03661320 | | **Trial Status** | Active, not recruiting | | **Sponsor Name** | Bristol Myers Squibb | | **Investigational Product** | Nivolumab, BMS-986205 (linrodostat mesylate), Gemcitabine, Cisplatin (Platinol) | | **Phase of Development** | Phase 3 | | **Medical Monitor** | Bristol Myers Squibb Clinical Operations Lead |

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To compare the pathological complete response (pCR) rate and event-free survival (EFS) of nivolumab plus neoadjuvant gemcitabine/cisplatin (GC) chemotherapy, followed by post-surgery continuation of immuno-oncology (IO) therapy, with neoadjuvant GC chemotherapy alone in adult participants with previously untreated muscle-invasive bladder cancer (MIBC). * **Secondary Objectives:** To evaluate Overall Survival (OS) and to describe the safety and tolerability (incidence of AEs and SAEs) of nivolumab and nivolumab/BMS-986205 in combination with GC chemotherapy.

2.2 Background & Rationale In cisplatin-eligible muscle-invasive bladder cancer (MIBC), cisplatin-based neoadjuvant chemotherapy (NAC) prior to radical cystectomy improves overall survival, but only a minority of patients achieve a pathologic complete response (pCR), and disease recurrence remains high. Tumor PD-L1 expression increases in MIBC after NAC, suggesting potential synergy in combining PD-1/PD-L1 inhibitors with NAC. Additionally, the IDO1 pathway is overexpressed in bladder cancer and is associated with poor outcomes. Combining the IDO1 inhibitor linrodostat mesylate (BMS-986205) with nivolumab and standard chemotherapy may significantly increase the rate of pCR and prolong EFS compared to chemotherapy alone.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Neoadjuvant chemotherapy alone) * **Blinding:** Double-Blind (for BMS-986205 vs. Placebo) and Open-label (for

Nivolumab and Chemotherapy) * **Randomization:** 1:1:1 Stratified Randomization * **Trial Status:** Active, not recruiting

3.2 Planned Enrollment & Duration * **Targeted Enrollment:** 855 participants * **Number of Sites:** Global (approximately 28 countries) * **Estimated Study Start:** 12-Oct-2018 * **Estimated Primary Completion:** 28-Nov-2023 (*Note: Overall study completion estimated for Dec-2026*)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Clinical stage T2-T4aN0M0 previously untreated Muscle-Invasive Bladder Cancer (MIBC). 2. Participant must be deemed eligible for Radical Cystectomy (RC) by his/her oncologist and/or urologist and must agree to undergo Radical Cystectomy (RC) after completion of neoadjuvant therapy. 3. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1.

4.2 Exclusion Criteria 1. Clinical evidence of positive lymph node(s) (LN) (≥ 10 mm in short axis) or metastatic bladder cancer. 2. Prior systemic therapy, radiation therapy, or surgery for bladder cancer other than transurethral resection of bladder tumor (TURBT) or biopsies is also not permitted. 3. Active, known, or suspected autoimmune disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * **Arm A (Control):** Gemcitabine & Cisplatin (Platinol) every 3 weeks for 4 cycles followed by Radical Cystectomy (RC). No adjuvant therapy. * **Arm B:** GC + Nivolumab (360 mg) every 3 weeks for 4 cycles + BMS-986205 Placebo (100 mg) daily $\rightarrow$ RC $\rightarrow$ Adjuvant Nivolumab (480 mg) every 4 weeks + BMS-986205 Placebo (100 mg) daily for 9 cycles. * **Arm C:** GC + Nivolumab (360 mg) every 3 weeks for 4 cycles + BMS-986205 (100 mg) daily $\rightarrow$ RC $\rightarrow$ Adjuvant Nivolumab (480 mg) every 4 weeks + BMS-986205 (100 mg) daily for 9 cycles. * **Route of Administration:** Intravenous (Nivolumab, Gemcitabine, Cisplatin) and Oral (BMS-986205/Placebo). * **Duration of Treatment:** Up to 12 weeks of neoadjuvant therapy, followed by surgery, and up to ~36 weeks (9 cycles) of post-surgery adjuvant therapy.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care during chemotherapy (e.g., anti-emetics, hydration). * **Prohibited:** Other systemic anticancer therapies, radiation therapy, and concurrent investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Imaging (CT/MRI), TURBT Pathology Review, Eligibility confirmation
Baseline	Day 0	Randomization, First dose of Neoadjuvant Treatment
Interim 1 (Neoadjuvant)	Every 3 Weeks	Safety Labs, Efficacy Assessment, Pre-surgery Imaging
Surgery	Within 6 Weeks post-NAC	Radial Cystectomy (RC) and Central Pathology Review
Interim 2 (Adjuvant)	Every 4 Weeks (Arms B & C)	Safety Labs, Post-surgery IO treatment administration
End of Study	Up to 5-7 Years	Survival follow-up, Long-term recurrence monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Pathological Complete Response (pCR) rate:** Assessed in all randomized participants, comparing Arm B vs. Arm A. * **Event-Free Survival (EFS):** Assessed in all randomized participants, comparing Arm B vs. Arm A.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) in all randomized participants. * Incidence of Adverse Events (AE) in participants who received at least one treatment dose. * Incidence of Serious Adverse Events (SAE) in participants who received at least one treatment dose. * Biomarker assessments including PD-L1 expression and gene mutation status.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Collected throughout the study per the NCI Common Terminology Criteria for Adverse Events (CTCAE) version 5.0 guidelines. Incidence is tracked for all participants who received at least one treatment dose.
- **Serious Adverse Events (SAEs):** Tracked for all participants receiving ≥ 1 treatment dose and must be reported to the sponsor within 24 hours of investigator awareness.
- **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (IDMC) will review safety data continuously during the trial.

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy outcomes (pCR and EFS). Safety Set for all AE/SAE evaluations (patients receiving ≥ 1 dose of study drug).
- **Sample Size Justification:** A targeted enrollment of 855 randomized patients provides adequate power to detect a

statistically significant improvement in EFS and pCR in experimental arms versus Arm A.

- **Handling of Missing Data:** For the pCR endpoint, participants who do not undergo surgery or lack pathological specimens are considered non-responders. EFS is censored at the date of last tumor assessment prior to the initiation of subsequent anticancer therapy.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Regular onsite and remote monitoring visits to evaluate protocol compliance and source data.
 - **Audit Trail:** Mandatory eCRF locking, source data verification (SDV), and robust data clarification form (DCF) workflows.
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11. Study Identifiers & Governance

- **Primary ID:** CA017-078
 - **Registry Number:** NCT03661320
 - **Sponsor/Lead Organization:** Bristol Myers Squibb
 - **Study Title:** A Study to Compare Chemotherapy Alone Versus Chemotherapy Plus Nivolumab or Nivolumab and BMS-986205, Followed by Continued Therapy After Surgery With Nivolumab or Nivolumab and BMS-986205 in Participants With Muscle Invasive Bladder Cancer
 - **Official Regulatory Title:** A Phase 3, Randomized, Study of Neoadjuvant Chemotherapy Alone Versus Neoadjuvant Chemotherapy Plus Nivolumab or Nivolumab and BMS-986205, Followed by Continued Post- Surgery Therapy With Nivolumab or Nivolumab and BMS-986205 in Participants With Muscle-Invasive Bladder Cancer
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12. Clinical Framework (Bladder Cancer Specific)

- **Disease Areas:** Urinary Bladder Neoplasms, Muscle-Invasive Bladder Cancer (MIBC)
 - **Diagnosis Threshold:** Clinical stage cT2-T4aN0M0
 - **Medical Methodology:** Neoadjuvant Systemic Therapy followed by Radical Cystectomy
 - **Optimization Target:** Pathological Complete Response (pCR) prior to surgery
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13. Study Procedures & Methodology

- **Management Pathway:** Neoadjuvant Chemotherapy (with/without Immunotherapy) $\rightarrow$ Radial Cystectomy $\rightarrow$ Adjuvant Immunotherapy
 - **Education Protocol (HCP):** Protocol training, immune-related adverse event (irAE) management, and surgical coordination
 - **Education Protocol (Patient):** Informed consent process, surgical preparation, and strict immune-mediated AE awareness/reporting
 - **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for EFS and Blinded Central Pathology Review for pCR
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 7 years (Follow-up weeks)
 - **Onsite Visit Cadence:** Every 3 weeks (Neoadjuvant Phase) & Every 4 weeks (Adjuvant Phase)
 - **Remote Monitoring Frequency:** Telephone/survival contacts structured throughout the follow-up period
 - **Checklist Review Interval:** Disease surveillance via CT/MRI every 12 weeks (Q12W) for the first two years, then every 6 months thereafter.
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15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				